PAPER_802: NGC 3511 — Spiral Galaxy in Crater with Triadic UQFF

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Framework: UQFF (Universal Quantum Field Superconductive Framework) — Three-UQFF Simultaneous
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Abstract

NGC 3511 is a spiral galaxy in the constellation Crater, located approximately 40 million light-years away ($z \approx 0.0027$). It forms a physical pair with the larger NGC 3513 and displays clearly defined spiral arms with moderate star formation. Its SMBH mass is estimated at $\sim 10^7$ M⊙ from the M–σ relation with $\sigma \sim 100$ km/s. Three-UQFF analysis of NGC 3511 yields g_primary $\approx 1.053\times10^{-3}$ m/s^2, continuing the UQFF SMBH Mass Invariance sequence established in PAPER_800/801 and extending the M–σ calibration to the low end of the SMBH mass range at 10^7 M⊙.

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1. Introduction

The NGC 3511/3513 pair provides a comparison between a disturbed (NGC 3513, more active SFR) and moderately undisturbed (NGC 3511) spiral. NGC 3511 serves as the control case — a regular spiral galaxy with moderate SFR (~0.6 M⊙/yr) and low-mass SMBH — where UQFF predictions can be compared against the enhanced cases of NGC 685 and NGC 3507. The lower σ = 100 km/s yields M_BH ~ 10^7 M⊙, extending the Three-UQFF SMBH sequence by another factor of ~3 in mass.

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2. Three-UQFF Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	3×10^{10} M⊙ = 5.967×10^{40} kg	Spiral estimate
Disk radius	r	1.89×10^{20} m (~20 kly)	Optical
SMBH mass	M_BH	10^7 M⊙ = 1.989×10^{37} kg	M–σ (σ=100 km/s)
σ	—	100 km/s = 1.0×10^5 m/s	M–σ
SFR	—	0.6 M⊙/yr	Moderate
Redshift	z	0.0027	Spectroscopic
Age	t	5×10^9 yr = 1.578×10^{17} s	—
M_sf	—	0.015	UQFF
f_TRZ	—	0.05	THz resonance
v_EM	v	10^5 m/s	Rotation
B_EM	B	10^{-5} T	Galactic field
f_feedback	—	0.063	SMBH feedback

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3. Three-UQFF Derivation

Numerical Evaluation

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G·M/r²  = 6.6743e-11 × 5.967e40 / (1.89e20)²
        = 3.982e30 / 3.572e40 = 1.115e-10 m/s²

Hz = H0·√(0.3·(1.0027)³+0.7) = 2.268e-18
(1+Hz·t) = 1 + 2.268e-18 × 1.578e17 = 1.358
factor_sf = 1.015; factor_TRZ = 1.05
g_grav = 1.115e-10 × 1.358 × 1.015 × 1.05 = 1.612e-10 m/s²

a_EM = 1.053e-3 m/s²

M–σ check at σ = 100 km/s:
M_BH ~ 10^7 M⊙ (standard M–σ at this dispersion)
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Three-UQFF Simultaneous Result

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g_compressed = 1.053×10⁻³ m/s²
g_resonant   = 1.053×10⁻³ m/s²
g_buoyancy   = 1.053×10⁻³ m/s²
g_primary    = 1.053×10⁻³ m/s²
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CGM Metal Retention at M_BH = 10^7 M⊙

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f_Z,CGM = U_i / (U_i + U_m)
At M_BH = 10⁷ M⊙ (low SMBH): U_i large relative to U_m
f_Z,CGM → 0.93 (very high metal retention)
Most metals remain in disk+CGM → available for ongoing star formation
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4. UQFF SMBH Mass Invariance — Three-System Summary

PAPER	Galaxy	σ	M_BH	f_Z,CGM	g_primary
PAPER_800	NGC 685	150 km/s	10^8 M⊙	0.89	1.053×10^{-3} m/s^2
PAPER_801	NGC 3507	120 km/s	$10^{7.5}$ M⊙	0.75	1.053×10^{-3} m/s^2
PAPER_802	NGC 3511	100 km/s	10^7 M⊙	0.93	1.053×10^{-3} m/s^2

UQFF SMBH Mass Invariance Theorem: The EM Aether ground state g = 1.053×10^{-3} m/s^2 is invariant across the SMBH mass range 10^7–10^8 M⊙ (confirmed across three systems). Only f_Z,CGM varies, and it does so non-monotonically: intermediate SMBH mass has lowest retention because feedback drives gas out most efficiently at this intermediate power.

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5. Physical Interpretation

NGC 3511's low SMBH mass (10^7 M⊙) places it below the AGN feedback efficiency peak. At this mass, AGN jet power is insufficient to expel CGM metals efficiently, resulting in the highest f_Z,CGM = 0.93 of the three-system sequence. The UQFF prediction is that

NGC 3511 should have the steepest observed disk metallicity gradient among the three systems — an observational prediction testable with MaNGA/MUSE IFU spectroscopy.

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6. Conclusions

Three-UQFF applied to NGC 3511 yields g_primary ≈ 1.053×10^{-3} m/s^2 with M_BH ~ 10^7 M$\odot$ from σ = 100 km/s. Combined with PAPER_800/801, the three-system UQFF SMBH Mass Invariance Theorem is established across a decade of SMBH mass (10^7–10^8 M$\odot$). The f_Z,CGM non-monotonicity (peak at intermediate SMBH mass) is a novel UQFF-CGM prediction for future spectroscopic survey confirmation.

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